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Magnetostratigraphic investigations on an Emeishan basalt section in western Guizhou province, China

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Abstract

A magnetostratigraphic investigation has been conducted on a late Permian Emeishan basalt section at Duge in western Guizhou province, China. The characteristic remanent magnetization (ChRM) directions resolved at high temperatures from the basalts pass the fold test with dual polarity and define a normal magnetozone of five sixths of the total thickness in the lower part and a reversed magnetozone of one sixth of the thickness in the top part of the section. The ChRM directions from the basalt clasts of pyroclastic rocks fail the conglomerate test, but the direction and polarity of the ChRM of the clasts from the upper reversed zone are consistent with those of the basaltic lavas in the reversed zone and the normal directions also dominate the ChRM of the clasts taken from the lower normal zone. This suggests that either the original remanence of the clasts was reset by cementing basaltic lavas or the ChRM of the basalts was acquired late but not beyond a field reversal. The magnetostratigraphic pattern (N–R) observed at Duge is as simple as that observed from some major continental flood basalt provinces in the world, where very few magnetic polarity zones were observed from volcanic sequences of great thickness; the emplacement of the Emeishan basalts in western Guizhou is estimated to be less than two million years. The normal polarity magnetozone observed indicates that the extrusion of the Emeishan basalts in western Guizhou postdates the Permo-Carboniferous reversed superchron (PCRS). The apparent shortness of the reversed magnetozone explains why far fewer reverse directions than normal ones have been so far reported from the Emeishan basalts. © 1998 Elsevier Science B.V. All rights reserved.

Keywords: magnetostratigraphy; basalts; Guizhou China; Yangtze Platform

1. Introduction

The late Permian Emeishan basalts in the western margin of the Yangtze Block (YB) are an important continental flood basalt (CFB) province which covers an area of more than 250 000 square kilometers in the border region between Sichuan, Yunnan and Guizhou provinces in southwestern China (Fig. 1) [1]. McElhinny et al. [2] conducted a pioneering paleomagnetic investigation at the Emei stratotype section that was followed by many later studies [3–9]. Through these studies, a general consensus about the assignment of the magnetic polarity has been reached that the predominant north–northeasterly shallow-up directions observed represent normal polarity and the west–southwesterly to southwesterly downward directions are of reversed polarity [7,10,11], yet some authors contend that the

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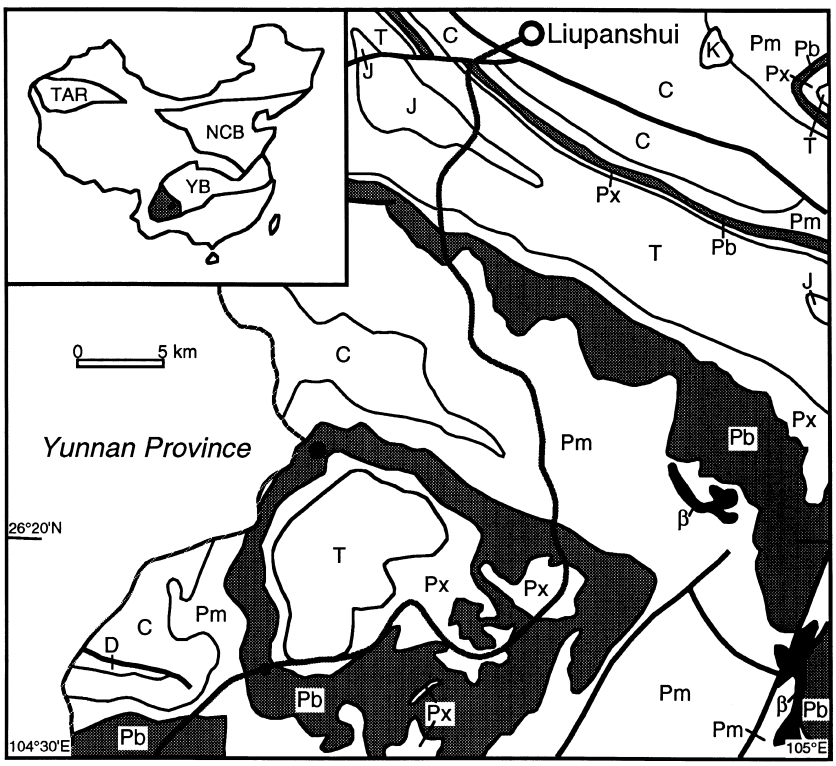


Fig. 1. Geologic sketch map of the sampling area. Solid circles denote the sampled sites. Thick lines are faults, and the shaded line is the highway from Liupanshui to Panxie. Strata: D = Devonian, C = Carboniferous, Pm = Upper Permian Maokou Fm, Pb = Emeishan basalts (shaded), Px = Upper Permian Xuanwei Fm, T = Triassic, J = Jurassic, K = Cretaceous. β is diabase sill. The inset shows three major cratons in China (NCB = North China Block, YB = Yangtze Block, TAR = Tarim Block) and the Emeishan basalt province in the southwestern Yangtze Block (shaded). The dot indicates the location of Duge.

former group of directions is a later remagnetization and only the latter group is primary because the two groups of directions lack antipodality [3,8,12]. This contention has important implications for the tectonic evolution of the YB since nearly all the Late Permian paleomagnetic poles for the YB are derived from the Emeishan basalts. However, most of the previous studies aimed at obtaining paleomagnetic poles, and some analyzed only a very small number of samples. Moreover, much fewer reversed directions than the normal ones have been so far reported, and the stratigraphic relation between the two polarities is unclear. Therefore, it has been suggested that magnetostratigraphic investigations on carefully chosen sections will help resolve this controversy [9,13]. This paper reports magnetostratigraphic data observed from an Emeishan basalt section in western Guizhou province.

2. Geology and sampling

Western Guizhou is in the eastern part of the Emeishan basalt province. The sampled section is near the village of Duge (26.4°N, 104.7°E) about 28 km south–southwest of Liupanshui City and located at the northern flank of a basin-like syncline (Fig. 1). The core of the syncline is composed of the Mid–Lower Triassic rocks and Upper Permian Xuanwei Formation, a terrestrial coal-bearing clastic sequence of latest Permian (Changxingian) age [14,15]. The Emeishan basalts and underlying early Late Permian Maokou limestone crop out on the flanks. The Regional Geological Survey of Guizhou Province previously included a 61-m-thick conglomerate bed into the top of the basaltic sequence at the Duge section. However, our observation indicates that the conglomerate sharply overlies a massive

aphyric basalt flow and grades upwards into lithic tuffites of the Xuanwei Formation. Furthermore, the natural remanent magnetization (NRM) of the samples drilled from the matrix and basalt clasts in the conglomerate is surprisingly as weak as limestones (see below). Therefore, we believe that it is more reasonable to regard this bed as the basal conglomerate of the Xuanwei Formation. Excluding this conglomerate, the total thickness of the basaltic sequence at the Duge section is 551 m. At least twelve flows can be recognized, with each flow usually starting with pyroclastics, followed by aphyric and/or (plagioclase-) porphyritic basalts, and ending up with amygdaloidal and/or vesicular basalts. The whole sequence is characterized by the presence of multiple beds of agglomerate and pyroclastic breccia in the lower part, and a predominance of massive basalt flows in the middle part, and the occurrence of tuff and tuffite in the upper part. This is similar to other Emeishan basalt sections in western Guizhou and apparently reflects variation in volcanisity with time [16].

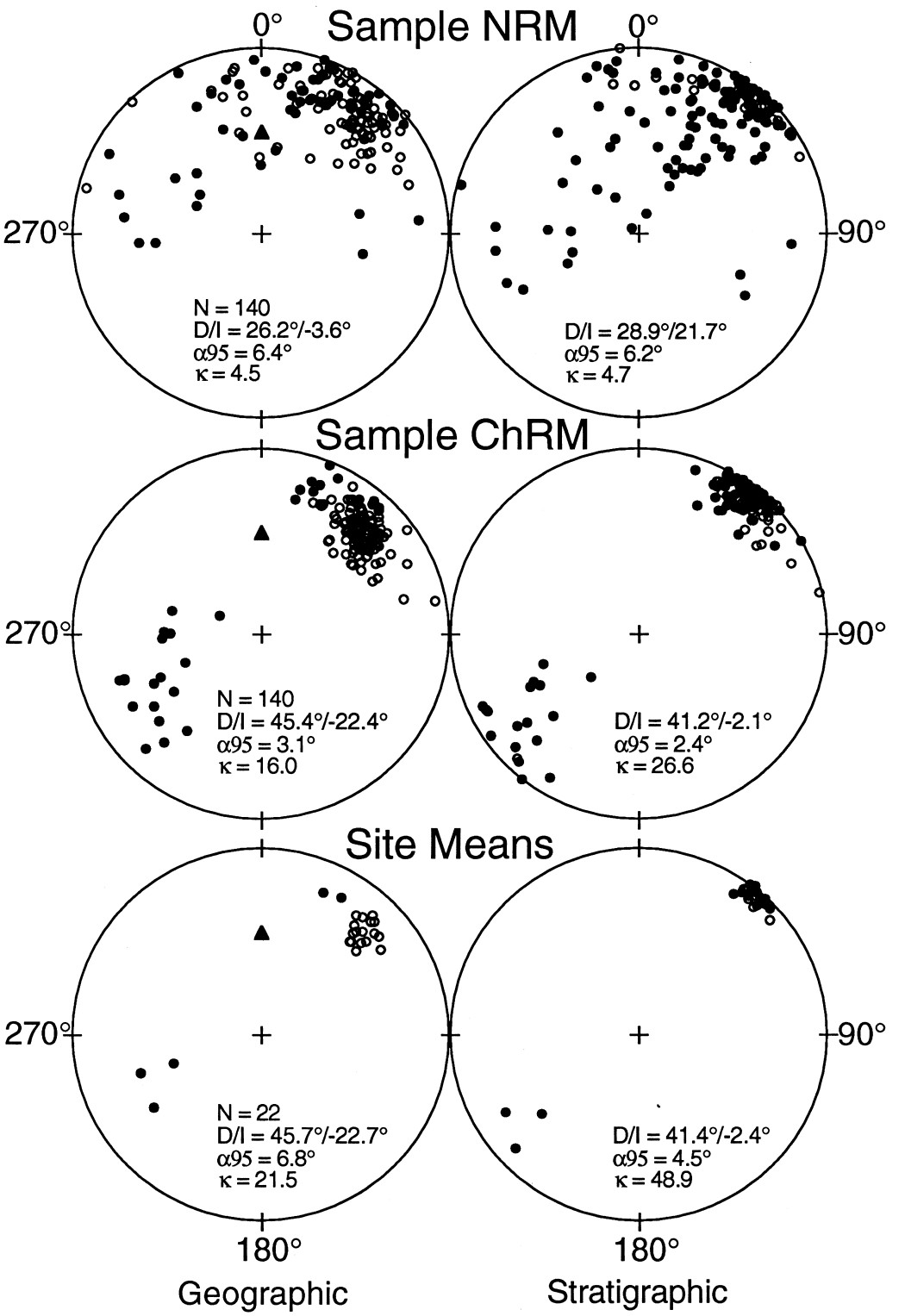
The Duge basalt section is exposed along a ravine. The lower and middle part was sampled along a newly excavated road that parallels the ravine, and the upper part was sampled from the bed of the ravine. In both cases, the exposures are very fresh. Cores were drilled about every 5 m using a portable gasoline-powered rock drill. A total of 140 samples were collected from the section. A pyroclastic breccia bed in the lower part and an agglomerate bed in the upper part of the basaltic sequence and the basal conglomerate of the overlying Xuanwei Formation were sampled for conglomerate tests. In addition, thirteen more samples were taken from the western flank of the syncline near a village called Jichang about 12 km south of Duge in order to conduct a fold test (Fig. 1). Bedding attitude can be accurately determined in the lower and upper part of the basaltic sequence thanks to the occurrence of the pyroclastic beds. The rocks dip nearly due south at 25–35° at Duge and dip east at 40° at Jichang, which conforms to the bedding attitude of overlying and underlying rock formations at the sampled sites. Both magnetic and sun compasses were used for core orientation whenever possible, and no significant difference was found between the two orientations.

3. Paleomagnetic results

All the samples were subjected to progressive thermal demagnetization using an ASC or Schonstedt thermal demagnetizer. Magnetization of the samples was measured in a 2G cryogenic magnetometer. These instruments are housed in a magnetically shielded room in the Paleomagnetic Laboratory at the University of Florida. Standard analytical techniques such as orthogonal vector plots [17], principal component analysis [18] and Fisherian statistics [19] were employed to analyze the paleomagnetic data.

3.1. Demagnetization

The NRM intensity of the basalt samples varies from 0.08 to 10.22 A/m, a typical range for basaltic rocks. The NRM directions are rather scattered (Fig. 2 upper), indicating that these samples are overprinted to various degrees. This was further confirmed by thermal demagnetization. Nearly all the samples have two magnetic components: a low temperature component (LTC), which accounts for 4–93% of the initial intensity (Fig. 3a, c, d and e), was generally removed below 400°C; a ChRM was resolved above 450°C from a majority of the samples and was commonly unblocked in the temperature range from 450 to 580°C (Fig. 3a, b, c, d and f), suggesting that magnetite is the magnetic carrier. ChRM directions were resolved from 109 out of 112 samples from the lower 460 m of the basaltic sequence at the Duge section and all of them are northeasterly with very shallow, mostly negative, inclinations in stratigraphic coordinates (Fig. 3a, b and c). ChRM directions were resolved from 18 out of 28 samples taken above this level and are all southwesterly and mostly downward (Fig. 3d, e and f). Clearly, the former group of directions represents the normal polarity and the latter represents the reversed polarity of the paleofield as described earlier. The thirteen samples collected at Jichang are all of normal polarity. The scatter in magnetic directions is dramatically reduced after thermal cleaning (compare Fig. 2 middle with upper). However, the reverse directions remain relatively scattered in comparison with the normal ones. This is mainly because the samples taken from the top part of the section are of-


Sample NRM
0°
270°
90°
N = 140
D/I = 26.2°/-3.6°
α95 = 6.4°
κ = 4.5
D/I = 28.9°/21.7°
α95 = 6.2°
κ = 4.7
Sample ChRM
N = 140
D/I = 45.4°/-22.4°
α95 = 3.1°
κ = 16.0
D/I = 41.2°/-2.1°
α95 = 2.4°
κ = 26.6
Site Means
N = 22
D/I = 45.7°/-22.7°
α95 = 6.8°
κ = 21.5
D/I = 41.4°/-2.4°
α95 = 4.5°
κ = 48.9
180°
Geographic
Stratigraphic

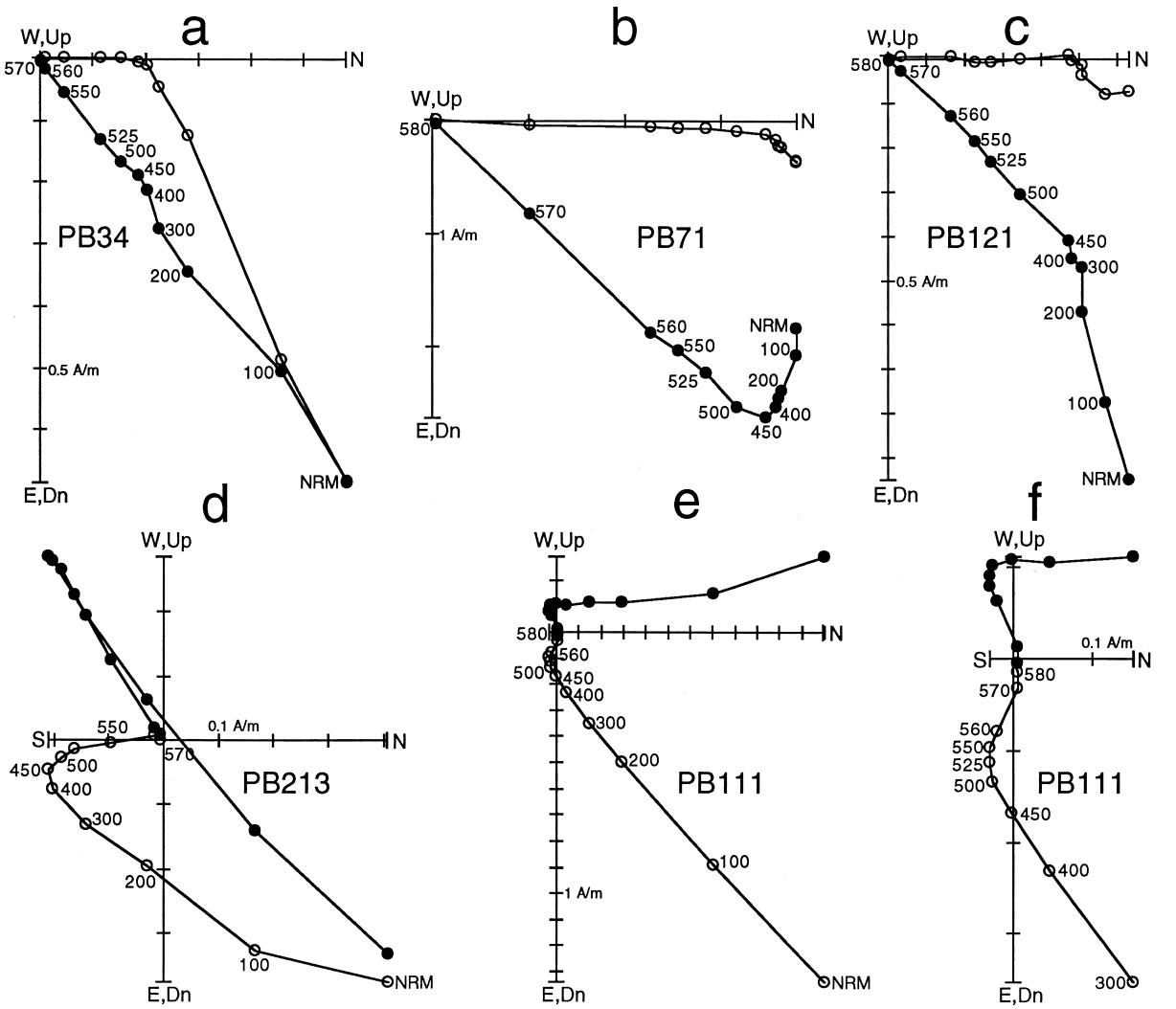


Fig. 3. Representative orthogonal plots of thermal demagnetization (in stratigraphic coordinates) for basalt samples. Solid (open) circles are projections of vector end points on horizontal (vertical) planes. Treatment levels are in ºC. (a), (b), (d) and (e) are from the Duge section on the northern flank of the syncline, (c) is from the western flank near Jichang. (f) is an partial enlargement of (e).

ten dominated by a northerly and steeply downward LTC which contributes a great proportion (30–94%) to the NRM and even the entire NRM (Fig. 3d, e and f). It is likely that thermal demagnetization did not completely eliminate the LTC in some of the reversed samples.

3.2. Fold test

As can be seen from Fig. 2 middle, tilt correction improves the grouping of the ChRM directions. In order to perform a standard fold test, six to eight contiguous samples through the sampled section are grouped into sites. Twenty sites are thus obtained

Fig. 2. Equal-area projections of sample NRM, ChRM and site mean directions from the Emeishan basalts. Solid (open) circles represent directions plotted onto the lower (upper) hemisphere. Black triangle denotes the present dipole field direction at Duge.

Table 1
Site and formation statistics for Emeishan basalts at Duge (26.4°N, 104.7°E)

Site	Bedding	n	R/N	In situ		Tilt corrected				Pole position	
				D (°)	I (°)	D (°)	I (°)	α_{95}	κ_s	Lat. (°N)	Long. (°E)
1	107/33	8	0/8	40.3	−24.2	38.3	6.5	5.8	93.3	46.6	220.4
2	89/35	6	0/6	49.9	−19.7	46.1	3.7	7.0	92.5	39.4	215.9
3	89/35	6	0/6	44.3	−17.6	42.2	7.8	8.2	68.3	43.8	216.5
4	89/35	6	0/6	48.0	−20.7	44.0	3.6	12.8	28.4	41.1	217.5
5	89/35	6	0/6	48.2	−33.3	38.7	−7.8	4.3	240.7	41.8	227.8
6	89/35	6	0/6	44.4	−31.7	36.4	−5.0	6.4	109.7	44.5	228.5
7	89/35	6	0/6	44.7	−24.5	39.7	1.5	9.9	46.7	44.0	222.0
8	89/35	6	0/6	42.2	−27.9	36.3	−0.7	9.4	52.2	46.0	226.3
9	89/35	7	0/7	47.5	−29.1	39.9	−3.8	5.6	117.4	42.2	224.7
10	89/35	6	0/6	43.4	−32.0	35.6	−5.0	8.0	71.2	45.1	229.2
11	89/35	6	0/6	42.6	−26.4	37.2	0.4	4.4	230.7	45.6	224.8
12	89/35	6	0/6	38.5	−20.2	36.0	7.6	7.7	76.4	48.8	221.7
13	88/25	7	0/7	40.7	−18.8	37.9	0.1	4.4	185.8	45.0	224.4
14	88/25	7	0/7	43.2	−14.3	41.5	3.7	9.8	38.5	43.2	219.3
15	88/25	6	0/6	45.2	−16.2	42.8	1.3	11.8	33.4	41.5	219.6
16	88/25	7	0/7	48.2	−26.8	42.1	−9.6	8.0	57.5	38.7	225.9
17	88/25	7	0/7	54.7	−23.2	48.9	−8.3	13.9	19.8	33.7	220.1
18	96/28	6	6/0	235.4	31.9	227.2	11.9	15.4	20.0	−34.0	43.1
19	96/28	6	6/0	251.6	50.0	230.6	33.5	11.7	43.7	−23.6	51.5
20	96/28	6	6/0	251.5	33.9	239.2	19.3	14.5	22.2	−22.0	38.8
21	18/40	6	0/6	30.4	16.2	37.8	4.6	8.9	57.6	46.5	221.9
22	18/40	7	0/7	23.5	18.5	34.1	10.6	8.6	49.3	51.3	221.6
All sites		22		45.7	−22.7	41.4	−2.4			41.4	222.8
				$\kappa_g = 21.5$		$\kappa_s = 48.9$		$\alpha_{95} = 4.5$		$A_{95} = 3.1$	
Normal		19		43.1	−20.0	39.8	0.6			43.7	222.5
				$\kappa_g = 27.8$		$\kappa_s = 133.5$		$\alpha_{95} = 2.9$		$A_{95} = 2.0$	
Reversed		3		245.7	38.8	232.3	21.6			−26.6	44.5
				$\kappa_g = 43.0$		$\kappa_s = 42.7$		$\alpha_{95} = 19.1$		$A_{95} = 13.4$	

Bedding: strike/dip; n: numbers of samples; R/N: numbers of reversed/normal polarity samples in a site; D, I: declination and inclination; α_{95}, A_{95}: radius of the 95% confidence circle about the mean direction and pole; κ_g, κ_s: Fisher precision parameters before and after tilt correction.

from the northern flank (Duge) and two from the western flank (Jichang). For these 22 sites, the precision parameter increases by a factor of two after tilt correction ($\kappa_s/\kappa_g = 48.9/21.5 = 2.27$); If the normal polarity sites alone are considered, improvement in grouping is even more striking ($\kappa_s/\kappa_g = 133.5/27.8 = 4.80$). Since the sites from the western flank are far fewer than those from the northern flank, it is better to use McFadden and Jones' fold test formulation [20] to see whether the two flanks share a common true mean direction before and after bedding correction. The statistic = 1.6551 ($R_1 = 21.0252$, $R_a = 19.6398$, $R_b = 1.9963$, $N = 22$) before unfolding, which is greater than the critical value (0.1616), so the hypothesis that the two flanks share a common true mean direction can be rejected at the 95% confidence level; by contrast, the statistic = 0.1105 ($R_2 = 21.5703$, $R_a = 19.6169$, $R_b = 1.9962$, $N = 22$) after bedding correction and the hypothesis of a common true mean direction after unfolding cannot be rejected. In addition, Incremental unfolding reveals that κ reaches maximum at 100% unfolding. It is concluded, therefore, that the fold test is positive and the ChRM was acquired prior to folding (Fig. 3 lower). Statistics are given in Table 1.

3.3. Reversal test

The antipodality of the normal and reverse directions was tested using the statistics given in Table 1.

The angular distance between the mean normal and reverse directions is 25.3 ± 22.0^{o} and the 95% confidence circles about them do not overlap. Applying McFadden and McElhinny's reversal test [21], this angle is greater than the critical angle (17.4°) at the 95% confidence level, so the reversal test is negative. The lack of antipodality between the normal and reverse directions commonly observed elsewhere in the Emeishan basalt province is thus duplicated.

3.4. Conglomerate test

Because the folding of the Permian rocks in the study area is believed to have taken place during early Cretaceous time [22], the positive fold test described above only demonstrates at best that the ChRM is pre-Cretaceous. To further constrain the age of the magnetization, the basal conglomerate bed of the Xuanwei Formation was first sampled for a conglomerate test (Fig. 6). This bed is 61 m thick with subangular to subrounded clasts cemented by fine-grained matrix. Composition of the clasts ranges from basalts to pyroclastics and clast size varies with most of them being smaller than 30×50 cm. Twenty-five samples were drilled from the basalt clasts and two samples from the matrix. Unfortunately, nearly all the samples, clast and matrix alike, are extremely weakly magnetized with the total NRM intensity mostly in the range from 0.02 to 0.16 mA/m; no ChRM directions were resolved. Initial thin section inspection has revealed pervasive alteration of plagioclase into sericite and zeolite, and pyroxene and glass into epidote, chlorite, and calcite in the clasts. Thus, no conclusion about the age of the magnetization can be drawn from this conglomerate bed.

An agglomerate was then sampled from the ninth flow from the bottom of the basaltic sequence (Fig. 6). This is a 6.5-m-thick massive basalt flow with local concentrations of clasts. The subangular to subrounded clasts are embedded in aphyric basalt and are mostly amygdaloidal basalt; clast size is around 10×30 cm. Sixteen samples were drilled from the clasts and two samples from the cementing basalt. The demagnetization behavior of the clast samples is very similar; all consist of two magnetic components: a LTC dominated the NRM and was removed below 300°C, and a ChRM was isolated above 500°C and often persisted up to 620°C (Fig. 4a, b and c). The two cementing basalt samples exhibited a single magnetic component which has the same direction as the LTC of the clast samples and was distributed in the temperature range from 100 to 570°C (Fig. 4d). The LTC directions of the clasts are well clustered with a mean of $D/I = 360.0^{o}/30.0^{o}$ in geographic coordinates (Fig. 4e). This mean direction is similar to a late Cretaceous overprint observed from the Triassic carbonates near Guiyang about 200 km east of Duge [23] and is possibly of the same origin. The tilt-corrected ChRM directions of the clasts are westerly to west–southwesterly with varying positive inclinations and are similar to the reverse ChRM directions observed from the basalts in the top part of the Duge section (Fig. 4f). Since the resultant, R (15.07), of the ChRM directions of the clasts is much greater than the 95% critical value, R_0 (6.40), for Watson's randomness test [24], these directions fail the conglomerate test at the 95% confidence level. Thus, the ChRM of the clasts was acquired no earlier than the formation of the agglomerate bed. Judging from the unblocking temperature spectra of these samples, the LTC is likely carried by magnetite and the ChRM is possibly carried by hematite or jointly by magnetite and hematite. The reddish coloration of the clasts supports this conclusion.

A pyroclastic breccia bed was also sampled from the second flow from the bottom of the basaltic sequence (Fig. 6). This bed is 5.7 m thick with angular to subangular clasts cemented by aphyric basalt. The clasts are chiefly amygdaloidal and aphyric basalt, mostly 1–4 cm in maximum dimension, and constitute 50–60% of the volume of the rock. Twenty-two samples were drilled from the clasts. Most of the clast samples possess a LTC which was eliminated below 200°C and may be carried by goethite (Fig. 5a and b). The LTC directions, although a bit scattered, fall close to the present dipole direction at Duge (Fig. 5d) and are possibly of recent origin. A ChRM was resolved from 19 samples which was discretely unblocked in the temperature range from 525 to 580°C (Fig. 5a, b and c), suggesting that the ChRM is carried by magnetite. It is also noteworthy that the ChRM direction persisted up to 600 or 610°C in some clasts (Fig. 5b and c), pointing to a small contribution from hematite or maghemite. According to Watson [24], the 95% critical value (R_0) for $N = 19$ is 6.98 which is less than the resultant

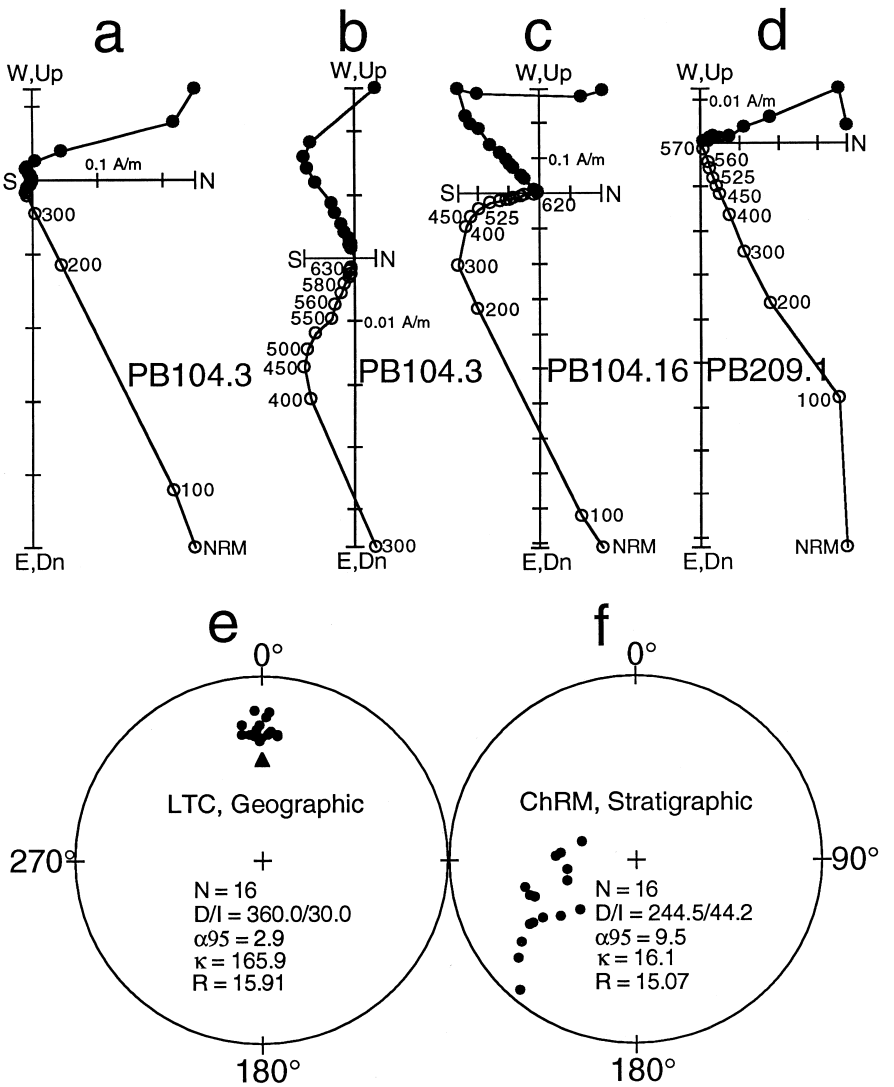


Fig. 4. Representative orthogonal plots (a, b, c, d) and equal-area projections of the LTC (e) and ChRM (f) directions of the basalt clasts from the agglomerate bed in the top part of the basaltic sequence. Coordinates and symbols as in Figs. 2 and 3.

($R = 8.64$) of the ChRM directions of the clasts, so the ChRM directions do not pass the conglomerate test at the 95% significance. In fact, seven clasts exhibited ChRM directions similar to the normal polarity ChRM directions observed from the basalts in the middle and lower part of the section (Fig. 5e). Nonetheless, the ChRM directions resolved from the clasts from this bed are more random than those from the agglomerate bed (Fig. 5e).

4. Discussion

4.1. Age of the magnetization

As described above, most of the basalt and basalt clast samples studied have two magnetic components. The LTC directions of the basalt samples are a bit scattered, but a large number of them fall close to the present Earth's magnetic field direction in geographic coordinates and have a mean of

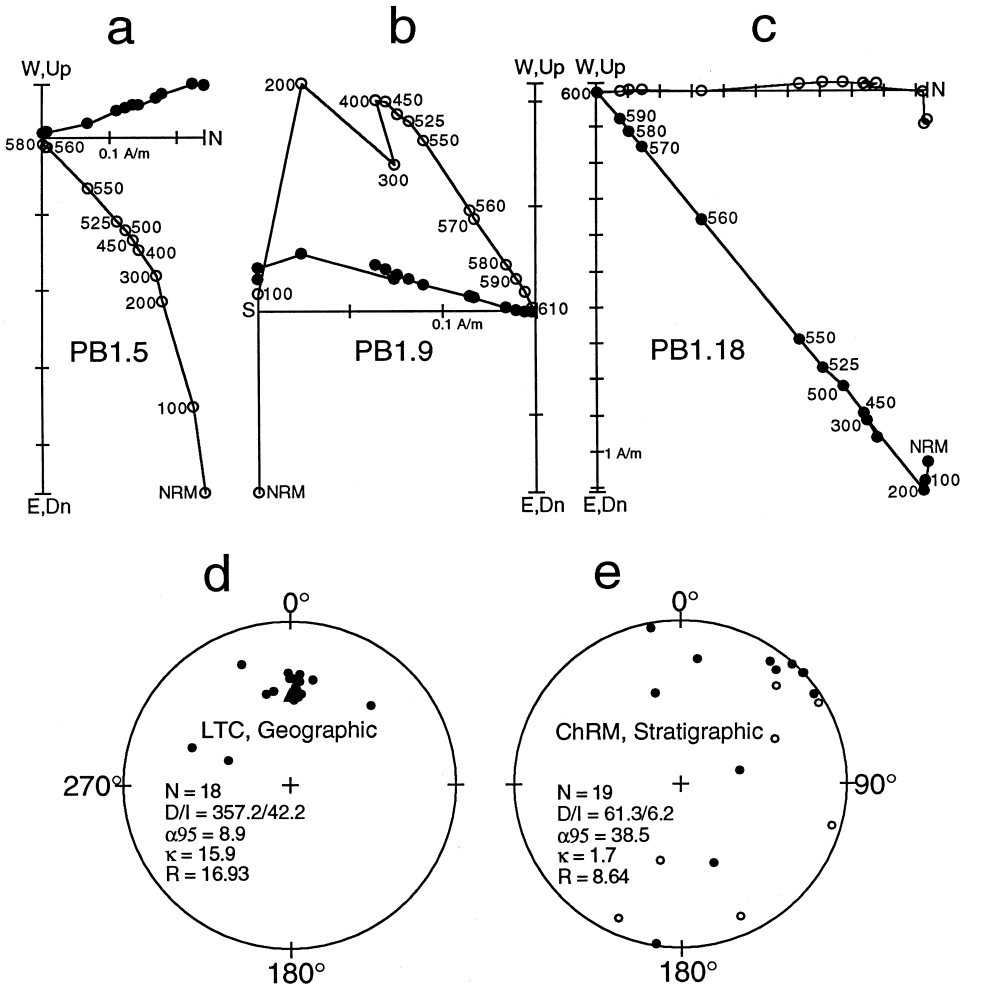


Fig. 5. Representative orthogonal plots (a, b, c) and equal-area projections of the LTC (d) and ChRM (e) directions of the basalt clasts from the pyroclastic breccia bed in the lower part of the basaltic sequence. Coordinates and symbols as in Figs. 2 and 3.

$D/I = 3.3^{o}/24.4^{o}$ ($\alpha_{95} = 6.6^{o}$, $\kappa = 4.2$, $N = 140$). This direction is similar to a magnetic overprint of recent to late Cretaceous age observed from the Triassic carbonates near Guiyang [23] and is likely of the same origin. The LTC data obtained from the basalt clasts in the agglomerate and pyroclastic breccia support such an explanation (Fig. 4d and Fig. 5d). The magnetic minerals which are responsible for the LTC cannot be ascertained without further experimentation, but possible candidates are magnetite and, to a lesser extent, goethite. The ChRM resolved at high temperatures from the basalt samples has been demonstrated to be prefolding and is thus presumably pre-Cretaceous in age. The unblocking temperature spectra indicate that magnetite is likely to be the magnetic carrier of the ChRM in these samples. However, the ChRM resolved from the basalt clasts either failed (in the case of the agglomerate bed) or did not statistically pass (in the case of the pyroclastic breccia) the conglomerate test, suggesting that acquisition of the ChRM in these clasts cannot be earlier than the emplacement of the lava flows which contain them. It is interesting, though, to note that the agglomerate bed, from which the basalt clasts exhibited reverse ChRM directions, occurs in the top part of the sequence which

shows exclusively reversed polarity. Likewise, the pyroclastic breccia bed occurs in the lower part of the sequence which is exclusively of normal polarity, and it is the normal ChRM directions displayed by some of the clasts that cause the clasts to fail the conglomerate test statistically. Also noteworthy is that, unlike the basalt samples, hematite or maghemite is a contributor to the ChRM of the basalt clasts. Therefore, it seems likely that the original remanent magnetization acquired by the basalt clasts during cooling was partially or completely reset through oxidation and reheating that took place during subaerial eruption and incorporation into basaltic lavas. The resetting, however, cannot be substantially later than the ChRM acquisition of the overlying lavas, otherwise the direction and polarity of the ChRM of the clasts might be different from that of the overlying basalts. In addition, the pole position derived from the ChRM of the basalts is significantly different from any post-Triassic pole so far reported from the YB. Thus, we believe that acquisition of the ChRM in the basalts is penecontemporaneous with the solidification of the lavas.

Some authors suggested that the normal polarity directions observed from the Emeishan basalts are a later remagnetization and only the reverse directions are primary [3,8,12]. The paleomagnetic data presented above are apparently at odds with such a contention. First, the reverse directions are only observed at the top part of the Duge section. Second, the reverse directions are more scattered than the normal ones, and some of them are apparently contaminated by the uncleaned LTC as shown by both the basalt and basalt clast samples. Third, the recovery rate of the reversed ChRM by demagnetization from the top part of the Duge section ($n/N = 18/28$) is much lower than the normal one from the lower part ($n/N = 109/112$), and field observation and initial thin section inspection reveal more alteration in the top part than in the lower part of the sequence. Therefore, at least at the Duge section, the normal polarity directions were acquired earlier and are less contaminated than the reversed ones.

4.2. Magnetic polarity sequence

Accepting that the ChRM resolved from the basalt samples from the Duge section is penecontemporaneous with the solidification of the lavas, the magnetic polarity sequence of the Emeishan basalts at this section is simple and straightforward (Fig. 6). The whole basaltic sequence consists of two magnetozones: a normal zone of about 460 m in the lower and middle part of the section and a reversed zone of 90 m at the top. Although the shallow inclinations are not particularly diagnostic, these two zones are well defined by the declination and virtual geomagnetic pole (VGP) latitude throughout the section. Similar simple structure of magnetic polarity stratigraphy has been observed from some major CFB provinces in the world [25–28]. For instance, the magnetostratigraphy of the Tertiary/Cretaceous Deccan traps in India has been interpreted to be composed of two normal and one reversed magnetic zones (N–R–N) with a large fraction of the volcanism having taken place during the middle reversed interval [25,27]. Likewise, a basaltic sequence of more than 3500 m in thickness in the Permian/Triassic Siberian traps was essentially laid down during one interval of normal magnetic polarity [29]. Such paleomagnetic observations in conjunction with ^{40}Ar–^{39}Ar age determination over the last decade have rendered a recognition that the bulk of a CFB is generated in rather short period of time on the order of one or two million years ([29] and references therein). Opdyke [30] has estimated that the frequency of reversal before and after the onset of the PCRS is similar to that observed before and after the Cretaceous quiet zone (KQZ). Assuming a frequency of one field reversal per million years, the duration of the eruption of the Emeishan basalts in western Guizhou would be less than 2 million years. Also, the emplacement of the Emeishan basalts appears to be post-PCRS at least at the Duge section. Zhou et al. [11] reported 12 reversed basalt samples from the basal part and 19 normal samples from above of an Emeishan basalt section at Ertan in southwestern Sichuan province. If this is true, the extrusion of the Emeishan basalts at Ertan might straddle the upper boundary of the PCRS. However, the Ertan section needs to be restudied because the original purpose of Zhou et al. was to derive paleomagnetic poles, and their sampling was not intensive enough to resolve the magnetostratigraphy. Finally, the reversed magnetozone is much shorter than the normal one at the Duge section, so the number of reversed samples

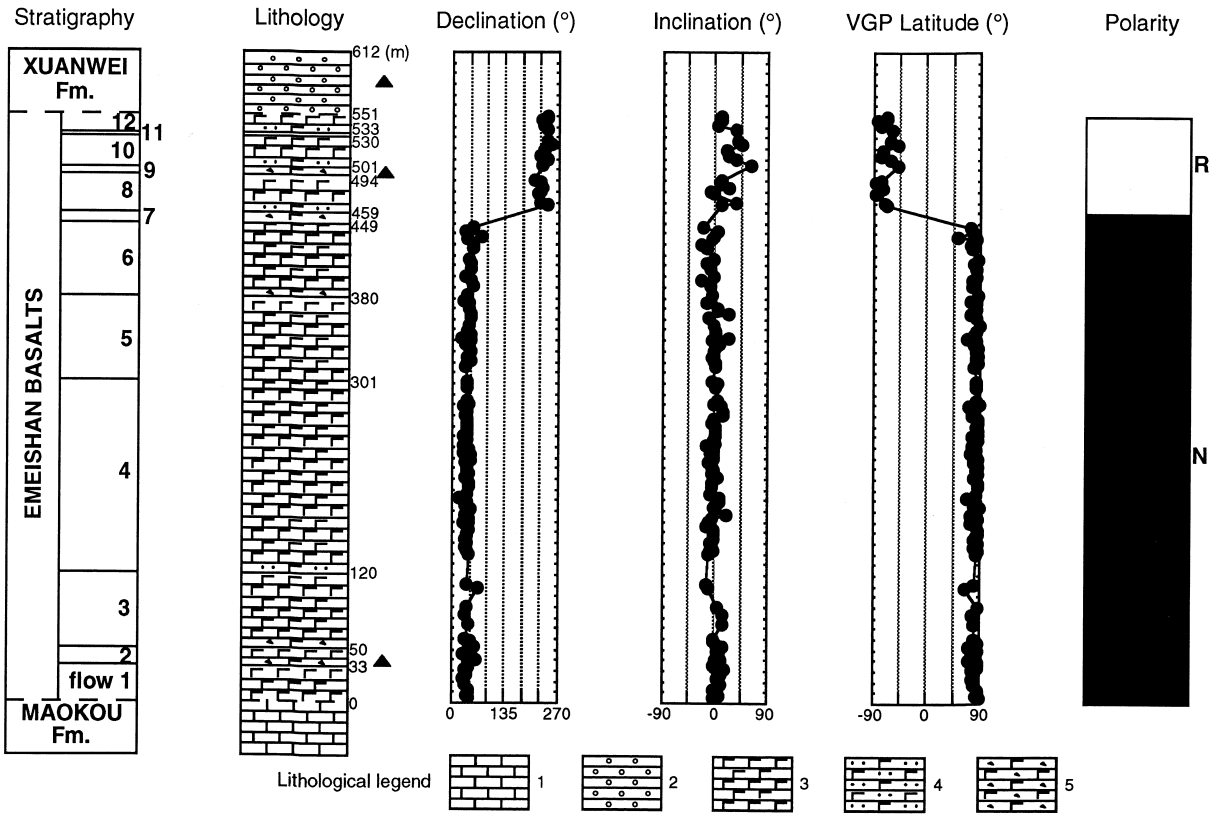


Fig. 6. Magnetostratigraphy and lithology of the Duge section with declination, inclination and VGP latitude plotted against stratigraphic thickness. Numbers next to the lithology column indicate stratigraphic thickness, and the black triangles denote the sampled conglomerate or pyroclastic beds. *R* and *N* next to the polarity column refer to reversed (white) and normal (black) magnetozones, respectively. Lithological legend: *1* = limestone, *2* = conglomerate, *3* = basalt, *4* = tuff/tuffite, and *5* = agglomerate/pyroclastic breccia.

observed is much smaller than that of the normal ones. This explains why very few reversed directions have been reported from the Emeishan basalts.

4.3. What might be the major cause for the difference between the paleomagnetic data sets from the Emeishan basalt province?

The paleomagnetic results so far reported from the Emeishan basalts are listed in Table 2 and plotted in Fig. 7. It is clear that the quality of individual studies varies widely in terms of number of samples analyzed. Nevertheless, it is apparent that significant difference in the mean directions exists between the data sets derived from different localities. Such difference is not only seen between normal and reverse directions but also between normal mean directions. As discussed above, the data observed from Duge do not support the contention that the lack of antipodality between the normal and reverse directions is because the latter is primary while the former is remagnetized. On the contrary, some of the reverse directions from Duge appear to be contaminated by uncleaned secondary magnetization. More work remains to be done before we can say that this is also true for other Emeishan basalt sections. For the normal directions, the inclinations observed from different localities are consistently shallow and the mean inclination is estimated to be $-7.3 \pm 9.3^{\circ}$ ($N = 9$) according to Enkin and Watson's method [31], so the directional difference is primarily due to declinational deviation. What then causes such declinational deviation between sampled localities? Considering the studies with analyzed samples more than 12

Table 2
Paleomagnetic results from the Emeishan basalts

Nr.	Localities	Coordinates	D_s/I_s	N/n	α_{95}	Paleopole	A_{95}	P	Reference
1	Emei	29.6°N, 103.4°E	16.4°/−4.5°	24/	3.5°	54.5°N, 254.3°E	3.9° [a]	N	[32]
2	Emei	29.6°N, 103.4°E	19.6°/−11.6°	/44	3.6°	49.7°N, 252.0°E	2.8°	N	[7]
3	Ereshan	29.7°N, 103.5°E	38.4°/−11.8°	/46	5.1°	38.7°N, 231.1°E	3.7° [a]	N	[5]
4	Huidong	26.7°N, 102.9°E	23.0°/−4.5°	/7	10.0°	53.5°N, 241.8°E	8.5°	N, R	[4]
5 [b]	Zhaojue	28.1°N, 102.9°E	18.1°/−4.9°	/19	7.0°	54.8°N, 250.3°E	4.9° [a]	N	[5]
6 [b]	Yanyuan	27.6°N, 101.9°E	7.8°/−3.9°	/17	10.7°	59.5°N, 266.4°E	7.6° [a]	N	[5]
7	Meigu	28.3°N, 103.0°E	17.9°/−5.8° [c]	/20	15.9° [c]	54.4°N, 251.2°E [c]	11.3° [a]	N	[6]
8	Ertan	26.5°N, 101.8°E	56.5°/−12.2° [c]	/19	8.8° [c]	26.3°N, 214.2°E [c]	6.4° [a]	N	[6]
9	Ertan	26.8°N, 101.8°E	234.5°/22.9°	/12	9.0°	24.3°N, 220.6°E	7.0° [a]	R	[6]
10	Dongchuan	26.1°N, 103.1°E	30.8°/5.7°	/8	25.0°	52.4°N, 225.9°E	16.5°	N, R	[4]
11	Binchuan	25.6°N, 100.6°E	242.0°/1.3°	3/9	24.8°	24.7°N, 204.3°E	18.1°	R	[4]
12	Kunming	25.0°N, 102.7°E	25.5°/−11.9°	10/57	6.1°	50.3°N, 241.0°E	4.5° [a]	N	[8]
13	Kunming	25.0°N, 102.7°E	244.1°/2.6°	1/2	–	22.7°N, 205.8°E	–	R	[8]
14	Zhijin	26.4°N, 105.7°E	224.5°/30.3°	1/5	12.0°	29.3°N, 235.3°E	9.9° [a]	R	[3]
15	Weining	26.7°N, 104.0°E	359.3°/−5.9°	1/9	10.0°	60.3°N, 285.4°E	7.1° [a]	N	[9]
16	Duge	26.4°N, 104.7°E	39.8°/0.6°	19/122	2.9°	43.7°N, 222.5°E	2.0°	N	This study
17	Duge	26.4°N, 104.7°E	232.3°/21.6°	3/18	19.1°	26.6°N, 224.5°E	13.4°	R	This study

Nr.: Numbers denoting directions in Fig. 7. N (n): Number of sites (samples). α_{95} (A_{95}): 95% confidence circles around mean directions (paleopoles). P: Polarity. [a] Estimated from dm and dp. [b] Directions have been checked with the original authors and paleopoles recalculated accordingly. [c] Recalculated from original data subsets. Note that some duplicated results from Mount Emei are not included.

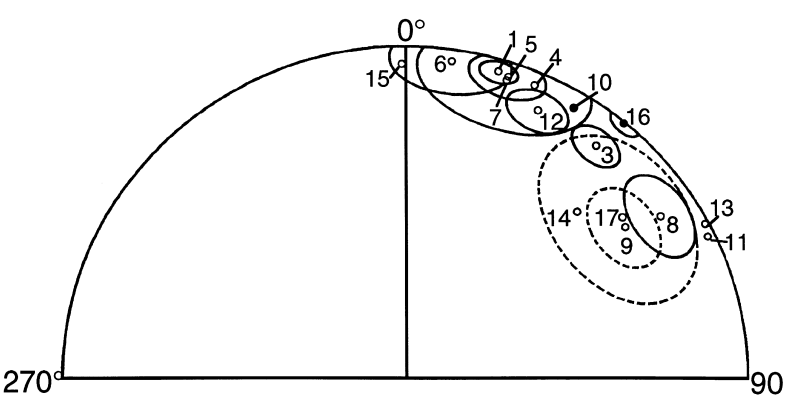


Fig. 7. Equal-area projections of the mean directions observed from the Emeishan basalt province. Numbers next to the circles are the same as in Table 2. Confidence circles are selectively indicated for the directions with more analyzed samples. Reverse mean directions are indicated by broken-lined confidence circles. Among the results derived from Mount Emei area, only those of Ma et al. [32] are indicated.

only, the normal and reverse directions observed at the same locality appear to be more consistent than the normal directions observed at different localities. For instance, the declinational deviation between the mean normal and reverse directions is $12.5 \pm 20.7°$ at Duge, W Guizhou and $2.0 \pm 13.3°$ at Ertan, SW Sichuan. However, the declinational difference between the mean normal directions observed at Ertan and Kunming is $31.0 \pm 10.9°$ and $14.3 \pm 6.8°$ between Duge and Kunming. Supposing that determination of the bedding attitude at these localities is accurate, the best explanation for such declinational deviation is rotation about vertical axis between the sampled localities even if some of the localities are very close to each other (e.g., the Ereshan section Liu et al. [5] sampled is only about 30 km away from the Emei type section Ma et al. [32] sampled and the declinational deviation between the two sections is

22.0 ± 6.3^{o}). It is unclear at present which localities were actually rotated and when. Wang and Van der Voo [33] and Wang et al. [34] reported a 'folding associated remagnetization' of possible Jurassic age from the Paleozoic rocks near Nanjing and from the Middle Triassic rocks from southern Guizhou. The mean directions of the remagnetization they reported are nearly north–south with very shallow inclinations and mixed polarities, and are similar to some northerly normal directions reported from the Emeishan basalts. Although remagnetization for such Emeishan basalt directions cannot be ruled out, more investigations in the YB need to be done to verify the results of Wang et al.

5. Conclusions

The ChRM directions resolved at high demagnetization temperatures from basalt samples collected from the Duge section in western Guizhou province are of dual polarity and pass the fold test but fail the conglomerate tests. Some of the reverse directions are apparently contaminated by an uncleaned secondary magnetization and result in non-antipodality between the normal and reverse directions. This is in contradiction with the previous suggestion that the normal polarity directions observed from the Emeishan basalts are remagnetized and only the reversed ones are primary. The lower five sixths of the section are exclusively of normal polarity whereas the top one sixth is of reversed polarity. The ChRM directions isolated from the basalt clasts of a basalt-cemented agglomerate bed in the reversed upper part are also reversed and the ChRM directions from the basalt clasts of a basalt-cemented breccia bed in the lower normal zone are likewise dominated by normal polarity. This seems to indicate that either the original magnetization of the clasts were reset by cementing lavas through reheating or magnetization of the basalts were delayed but did not exceed a field reversal. The simple N–R magnetostratigraphy of the Emeishan basalts observed at Duge is very similar to that reported from some major CFB provinces in the world such as the Deccan traps in India and the Siberian traps in Russia, and the emplacement of the basaltic sequence in western Guizhou is estimated to be less than 2 Myr. The very short reversed magnetozone can also explain the scarcity of reverse directions reported from the Emeishan basalts. The normal and reverse directions observed from the same section appear to be more consistent than the normal directions derived from different sections in the Emeishan basalt province, so rotation about vertical axis between different localities is probably the major cause responsible for such significant deviation. More sections need to be investigated in the future to see if similar magnetic polarity sequence can be found. If correlatable magnetostratigraphy can be established on multiple sections across the Emeishan basalt province, it will be another line of evidence for the primary origin of the ChRM observed.

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